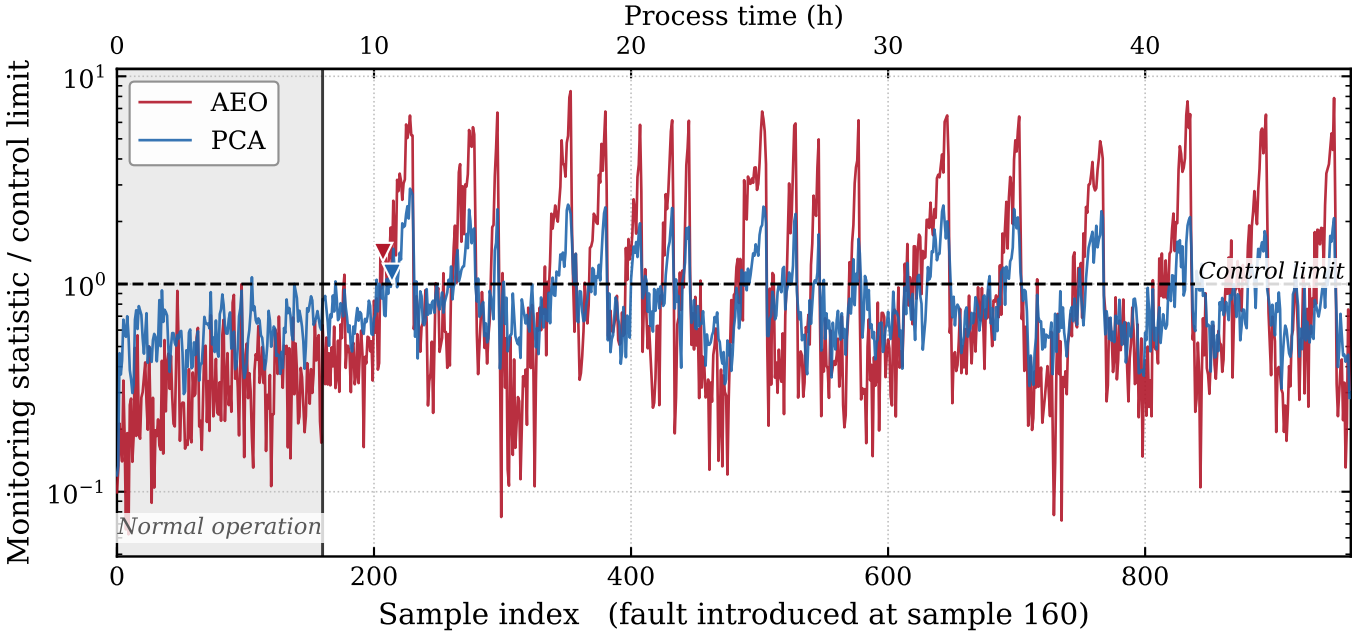
Fault Detection Performance for IDV(20)

Hotelling's T^2 statistic — AEO vs. PCA at a common 1% false-alarm rate



Held-out performance (mean of 7 independent runs)

AEO	delay	51.3 samples (154 min)	MDR 0.654	detected in 7/7 runs
PCA	delay	55.4 samples (166 min)	MDR 0.770	detected in 7/7 runs

AEO detects 4.1 samples (12 min) earlier; AEO missed-detection rate lower by 15%